DOCUMENT C | SOLUTIONS 4

Worked solutions and proposed marking scheme

Full-syllabus coverage practice paper. These are independent suggested answers and mark allocations, not official university solutions.

Award method marks for correct reasoning and units even when a later arithmetic slip occurs; avoid penalizing the same carried error repeatedly. Equivalent physically correct explanations and valid stated assumptions can earn credit. A part's marking points total its question allocation. Diagrams are schematic and not to scale. The complete question text is repeated below so each solution file is usable independently.

Source notation

S1-S06 refer to the supplied six lecture decks and PDF page/slide numbers. A1/A2 are the supplied thermal/OCT student assignments, cited by PDF page including covers. "Recurring family" identifies the historical concept, not copied wording; original extensions are labeled. A historical reference means the closest related question, not evidence that the exact model task appeared.

Equation corrections

Positive activity is -dN/dt. Conventional spin echoes use intrinsic T2 in the ideal echo envelope, whereas gradient echoes retain T2* sensitivity. RF bandwidth means full width unless explicitly stated otherwise. All calculations use γ=267.52×10^6 rad s^-1 T^-1 and unrounded intermediate values.

Model 4 | Question 1: Sampling, image processing and nuclear instrumentation

1(a) [5 marks]

Question. An ideal 10-bit rounding ADC has input range 0-2.048 V and uses 1024 equal-width bins. Calculate its code width and maximum quantization-error magnitude. A useful input signal spans only 0-0.128 V; propose a gain to use the full range and explain the risk. Give three ADC specifications relevant to imaging.

Expected answer and working. Bin width=2.048/1024=0.002 V=2 mV; maximum rounding error is half a bin, 1 mV. Nominal gain=2.048/0.128=16, but practical gain should leave headroom because an unexpectedly larger input clips. Relevant specifications include bit depth/dynamic range, permitted input voltage range and maximum sampling rate (or usable input bandwidth). Bin endpoints are defined in the question, avoiding the 2^n versus 2^n-1 convention ambiguity.

Proposed allocation: 2 mV width (1); 1 mV error bound (1); Gain 16 (1); Clipping/headroom (1); Three relevant specifications (1). **Total 5.**

Source map: S1 pp.62-68. **Closest past question:** 2023 Q1(b)i-iii. **Status:** Original numerical variant.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(b) [4 marks]

Question. A real low-pass analogue signal contains frequencies up to 18 kHz. State the theoretical Nyquist sampling bound and explain the role of an analogue anti-alias filter. A 16-fold oversampling/decimation scheme is then used; what ideal equivalent-bit improvement is suggested by the lecture, and under what noise assumption?

Expected answer and working. The theoretical bound is $f_s \geq 2 \cdot 18 = 36$ ksample/s; a practical design uses margin and an appropriate filter transition band. The analogue anti-alias filter suppresses out-of-band components before sampling so they cannot fold into the signal band. The lecture rule is one effective bit per fourfold oversampling, hence $16 = 4^2$ gives +2 bits after low-pass filtering/decimation. It assumes adequately uncorrelated/noise-like quantization errors; oversampling alone cannot guarantee extra bits for a deterministic undithered error.

Proposed allocation: 36 ksample/s bound (1); Pre-ADC filtering / alias explanation (1); +2 bits (1); Noise assumption / decimation condition (1). **Total 4.**

Source map: S1 pp.67-72. **Closest past question:** No direct historical sampling numerical. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(c) [3 marks]

Question. Compare spatial low-pass and high-pass filtering in terms of detail, noise and diagnostic visibility.

Expected answer and working. Low-pass filtering suppresses high spatial frequencies, reducing noise but smoothing edges and small features. High-pass filtering emphasizes rapid spatial changes and edges but can increase visible noise. Neither guarantees greater diagnostic utility: detectability depends on lesion contrast, resolution and noise together, and processing cannot restore genuinely absent measurements without assumptions.

Proposed allocation: Low-pass effect and trade-off (1); High-pass effect and trade-off (1); Visibility depends on task / information limit (1). **Total 3.**

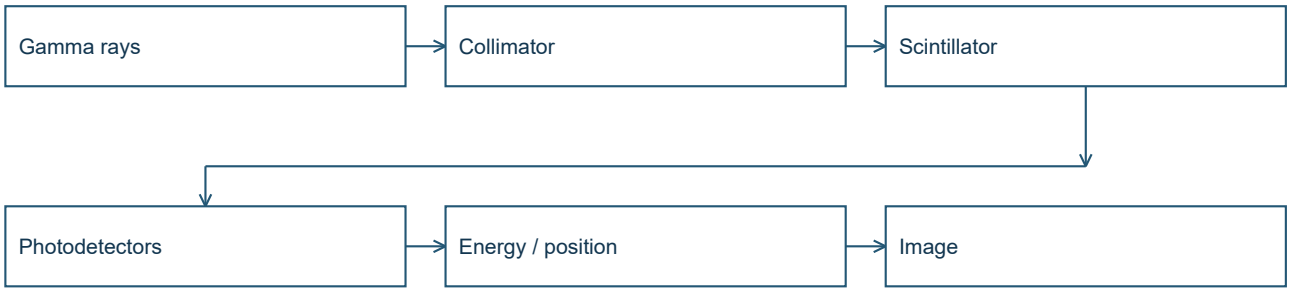
Source map: S1 pp.33-37,73-74. **Closest past question:** No direct historical filtering question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(d) [5 marks]

Question. Draw the signal path of a scintillation gamma camera. Explain the roles of collimator, scintillator, photodetectors, position/energy electronics and energy-window selection.

Expected answer and working. The collimator admits a restricted set of photon directions at a major sensitivity cost. Absorption in a scintillator creates light; photodetectors convert and amplify that light. Relative photodetector outputs locate the event, and their total/pulse height estimates deposited energy. An energy window around the photopeak rejects many scattered lower-energy events. Accepted positions accumulate into an image of tracer distribution. The chain is gamma photon → collimator → scintillator → photodetectors → position/energy logic → energy acceptance → image.



Proposed allocation: Collimator direction selection (1); Scintillator conversion (1); Photodetection (1); Position/energy determination (1); Window selection and labeled chain (1). **Total 5.**

Source map: S06 pp.11-13,21,35-37. **Closest past question:** 2024 Q1(b)i is closest SPECT context. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(e) [4 marks]

Question. Explain why SPECT needs scatter and attenuation correction. State how a combined SPECT/CT system assists quantitative interpretation.

Expected answer and working. Scatter changes photon direction and can deposit a count at an incorrect location; energy-window measurements help estimate/reject its contribution. Attenuation reduces counts more strongly for photons from deeper locations, so raw image brightness is not proportional to local activity alone. CT supplies a patient-specific attenuation map, appropriately converted to the gamma energy, and anatomical localization for the functional SPECT distribution. Registration errors can still impair the fusion.

Proposed allocation: Scatter mechanism (1); Energy-window approach (1); Depth-dependent attenuation (1); CT attenuation/localization benefit (1). **Total 4.**

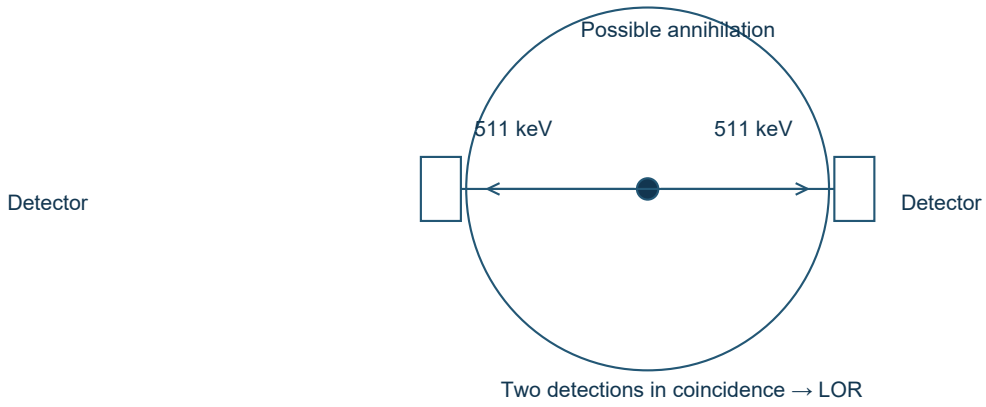
Source map: S06 pp.20-24. **Closest past question:** 2024 Q1(b)ii is closest SPECT/PET comparison. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

1(f) [4 marks]

Question. Using a simple sketch, distinguish a PET line of response from the position of an annihilation event. Explain why the detection of one coincidence does not by itself create a complete image.

Expected answer and working. Two opposing detectors define a chord/LOR containing the annihilation position under ideal unscattered geometry. Ordinary coincidence establishes that the pair belongs together but not a unique position along the chord. Multiple events and many LOR orientations constrain the tracer distribution tomographically. Counting noise, scattering, random events, positron range and detector dimensions limit the estimate; a diagram should mark a possible event point somewhere on the LOR.



Proposed allocation: Two detectors and LOR sketch (1); Event can lie along LOR (1); Many LORs and reconstruction needed (1); Physical limitation / count statistics (1). **Total 4.**

Source map: S06 pp.25-34. **Closest past question:** 2025 Q1(b)ii. **Status:** Recurring concept with additional reasoning.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 4 | Question 2: Radiography and CT beyond the common repeats

2(a) [4 marks]

Question. Explain the origin of the continuous and characteristic components of an X-ray-tube spectrum. State the different primary effects of increasing kVp and increasing mAs.

Expected answer and working. Decelerating electrons in the target nuclear field emit a continuous bremsstrahlung spectrum. Inner-shell vacancies followed by electronic transitions emit discrete characteristic lines specific to target material. Increasing kVp raises maximum photon energy and generally changes beam output/penetration; increasing mAs primarily raises photon number for a given kVp without setting a new maximum energy. Most tube electron energy becomes heat.

Proposed allocation: Bremsstrahlung (1); Characteristic transitions (1); kVp effect (1); mAs effect (1). **Total 4.**

Source map: S2 pp.7,13-15. **Closest past question:** 2025 Q2(a)i is closest tube question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(b) [4 marks]

Question. Compare photoelectric absorption and Compton scattering for diagnostic X-rays. Explain why pair production is not significant at a maximum tube photon energy of 120 keV.

Expected answer and working. Photoelectric absorption removes the incident photon and ejects a bound electron; its strong dependence on photon energy and effective atomic number contributes to tissue/contrast-agent differences. Compton interaction ejects a loosely bound electron and redirects a lower-energy photon, adding scatter that reduces image contrast. Both can deposit energy. Pair production requires at least approximately 1.022 MeV, well above 120 keV=0.120 MeV, so it is energetically impossible for that beam.

Proposed allocation: Photoelectric mechanism (1); Contrast dependence (1); Compton mechanism and scatter effect (1); Threshold comparison (1). **Total 4.**

Source map: S2 pp.17-25. **Closest past question:** 2024 Q2(a)ii. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(c) [4 marks]

Question. Compare computed radiography (CR), indirect digital radiography and direct digital radiography by their signal-conversion and readout principles.

Expected answer and working. CR stores a latent image in a photostimulable phosphor plate and uses a laser reader to release light for digitization. Indirect DR converts incident X-rays to light in a scintillator, then to charge in a photodiode/flat-panel array. Direct DR converts X-rays to charge in a photoconductor and reads the charge electronically. Digital output does not imply that the primary conversion is direct; each approach has its own resolution and efficiency trade-offs.

Proposed allocation: CR storage and laser readout (2); Indirect X-ray→light→charge (1); Direct X-ray→charge (1). **Total 4.**

Source map: S2 pp.39-42. **Closest past question:** 2025 Q2(a)i is closest detector context. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(d) [5 marks]

Question. A mass of 0.20 kg absorbs 0.60 mJ of X-ray energy. Calculate absorbed dose in Gy. With radiation weighting factor wR=1, determine equivalent dose in Sv. If only this tissue is exposed and its tissue weighting factor is 0.12, calculate its contribution to effective dose. Distinguish these three quantities.

Expected answer and working. Absorbed dose D=energy/mass=(0.60e-3)/0.20=0.003 J/kg=0.003 Gy=3 mGy. Equivalent dose H=wR D=0.003 Sv=3 mSv for the given radiation weighting. Effective-dose contribution E=wT H=0.12·0.003=0.00036 Sv=0.36 mSv. Absorbed dose is physical energy per mass; equivalent dose weights radiation type; effective dose weights irradiated tissue contributions. Equal numerical values for X-rays do not make Gy and Sv the same physical quantity.

Proposed allocation: Energy conversion and D=3 mGy (2); H=3 mSv (1); E contribution=0.36 mSv (1); Distinguish physical and weighted quantities (1). **Total 5.**

Source map: S2 pp.54-59. **Closest past question:** No direct historical dose numerical. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(e) [4 marks]

Question. For this question define multi-detector helical pitch as table travel per rotation divided by total active collimated beam width. A scanner has 16 active rows each 1.25 mm wide, table travel 30 mm/rotation and rotation time 0.50 s. Calculate pitch and table speed; neglect end effects and find time to cover 300 mm. Explain why the pitch convention must be stated.

Expected answer and working. Total active width=16·1.25=20 mm. Pitch=30/20=1.5. Table speed=30/0.50=60 mm/s; covering 300 mm takes 300/60=5.0 s, excluding extra reconstruction/end coverage. A single-slice convention can use one collimated slice thickness; substituting 1.25 mm here would produce 24 and is not the specified multi-detector convention. The lecture gives a single-slice pitch formula, so the extension is explicitly supplied.

Proposed allocation: 20 mm width (1); Pitch 1.5 (1); 60 mm/s and 5 s (1); Convention explanation (1). **Total 4.**

Source map: S3 pp.39-41; extended pitch definition supplied in question. **Closest past question:** 2023 Q2(b)ii is closest scanner-geometry precedent. **Status:** Original syllabus application.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

2(f) [4 marks]

Question. Explain the causes of (i) a ring artifact in third-generation CT and (ii) cupping in a uniform phantom. Give one corrective action for each.

Expected answer and working. A persistently miscalibrated detector samples a consistent radial position as the tube/detector arc rotates, producing a ring after reconstruction; recalibration or detector service addresses it. Cupping can arise from beam hardening: lower-energy photons are preferentially absorbed and the centre attenuation is underestimated under a simple monoenergetic model. Beam-hardening calibration/correction or appropriate filtration can reduce it; scatter correction also addresses scatter-related cupping.

Proposed allocation: Ring mechanism (1); Detector calibration/service (1); Cupping mechanism (1); Appropriate correction (1). **Total 4.**

Source map: S3 pp.56-59. **Closest past question:** 2023 Q2(b)iii is closest QA context. **Status:** Original syllabus coverage.

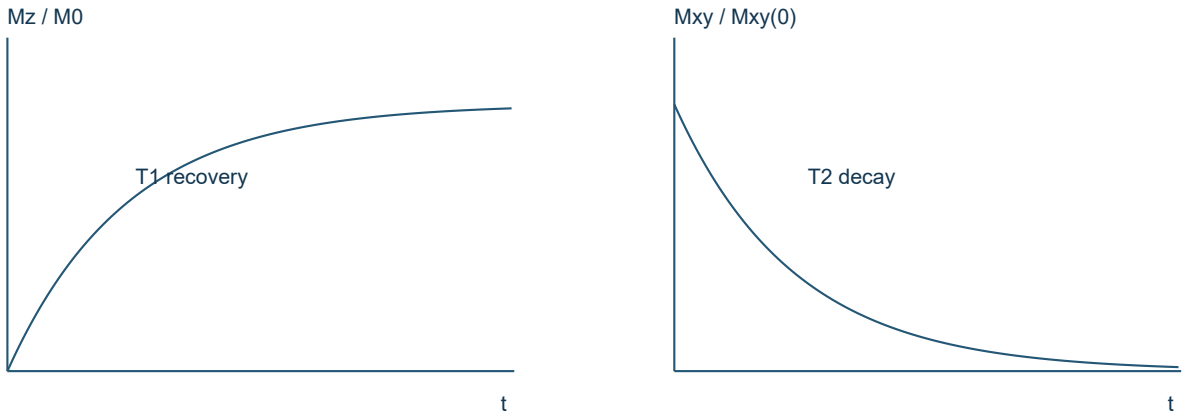
Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 4 | Question 3: MRI relaxation, reconstruction and functional imaging

3(a) [4 marks]

Question. After a 90° excitation, write and sketch ideal longitudinal recovery and transverse decay. Distinguish T1, T2 and T2*.

Expected answer and working. For initially saturated longitudinal magnetization Mz(0)=0, Mz(t)=M0[1-exp(-t/T1)]. The transverse envelope is Mxy(t)=Mxy(0)exp(-t/T2) when only intrinsic dephasing is considered. At t=T1, longitudinal recovery is 63.2% of M0; at t=T2, transverse signal is 36.8% of its initial value. T1 concerns spin-lattice recovery; T2 describes irreversible transverse coherence loss; T2* additionally includes reversible static-field dephasing and is no longer than T2 under the usual model.



Proposed allocation: Longitudinal equation/sketch (1); Transverse equation/sketch (1); T1 vs T2 interpretation (1); T2* distinction (1). **Total 4.**

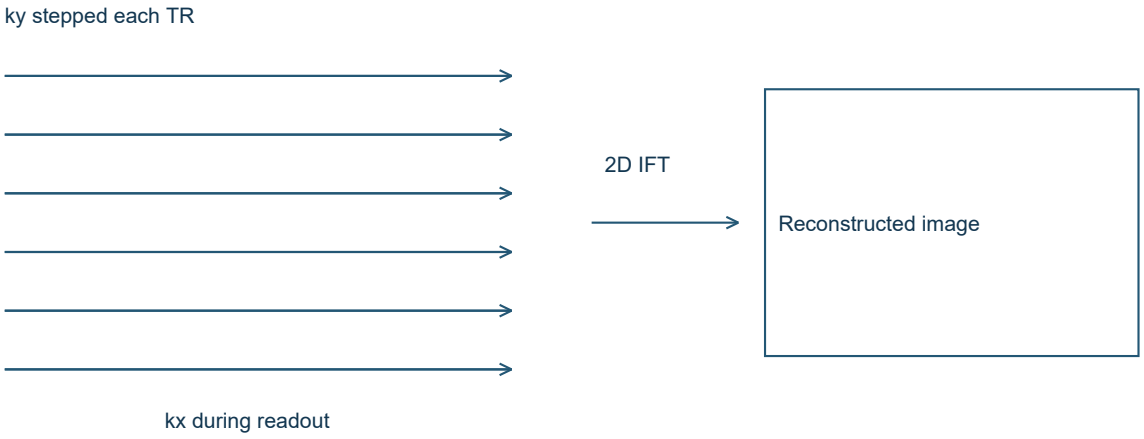
Source map: S4 pp.29-43. **Closest past question:** 2025 Q3(b)v-vi is closest relaxation mechanism. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(b) [5 marks]

Question. Explain how phase- and frequency-encoding gradients sample a 2D k-space grid, how the image is reconstructed, and how reducing Δk or increasing the maximum sampled spatial frequency changes the image.

Expected answer and working. A phase-gradient area sets ky for a repetition; its value is stepped across repeated TRs. The read gradient makes phase accumulate continuously during ADC sampling so kx is traversed along a line. The signal samples a spatial Fourier representation of the slice; a 2D inverse Fourier transform yields the image. FOV=1/Δk, so smaller sample spacing increases field of view. Extending the sampled spatial-frequency range (larger |kmax|) supports finer detail; this is distinct from merely interpolating a grid.



Proposed allocation: Phase gradient sets ky (1); Read gradient traverses kx (1); Inverse Fourier reconstruction (1); Smaller Δk gives larger FOV (1); Larger |kmax| supports finer resolution (1). **Total 5.**

Source map: S4 pp.57-71,87-90. **Closest past question:** 2025 Q3(b)i-iii is closest encoding precedent. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(c) [4 marks]

Question. State suitable qualitative TR and TE choices for proton-density-weighted, T1-weighted and T2-weighted conventional spin-echo images. Explain why long TR reduces T1 contrast.

Expected answer and working. Proton-density weighting uses long TR and short TE to minimize both relaxation weightings. T1 weighting uses relatively short TR and short TE so differential recovery remains. T2 weighting uses long TR and relatively long TE so transverse decay dominates. Long TR allows tissues to approach full longitudinal recovery, making (1-exp(-TR/T1)) approach 1 for both and reducing dependence on T1. Choices are relative to tissue relaxation times, not fixed universal millisecond thresholds.

Proposed allocation: PD long TR/short TE (1); T1 short TR/short TE (1); T2 long TR/long TE (1); Recovery explanation (1). **Total 4.**

Source map: S4 pp.75-81. **Closest past question:** 2024 Q3(a)i is closest intensity question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(d) [3 marks]

Question. An object extends beyond the encoded field of view in the phase direction. Explain wraparound aliasing and give two acquisition remedies with their trade-offs.

Expected answer and working. Discrete k-space sampling makes the reconstructed image periodic with period FOV=1/Δky. Signal outside the chosen FOV folds into the displayed region. Increase phase FOV by reducing Δky, with enough additional phase steps to preserve resolution (longer scan), or use phase oversampling with later cropping (more data/time). Spatially selective excitation/saturation is another acceptable remedy when applicable. Merely adding interpolated display pixels does not remove true aliasing.

Proposed allocation: Periodic sampling / folding explanation (1); Valid remedy with trade-off (1); Second valid remedy with trade-off (1). **Total 3.**

Source map: S4 pp.87-91. **Closest past question:** No direct historical aliasing question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(e) [5 marks]

Question. Explain the BOLD basis of fMRI, why rapid EPI acquisition is useful, and how a simple repeated task/rest experiment identifies associated regions. State one interpretation limitation.

Expected answer and working. Neuronal activation changes local haemodynamics; increased oxygenated blood flow can reduce local deoxyhaemoglobin concentration. Deoxyhaemoglobin is paramagnetic, so the changed susceptibility alters transverse dephasing and commonly increases T2*-weighted GE signal in activated regions. EPI provides rapid repeated whole-brain sampling. Record repeated task/rest blocks and test voxel signal association with the expected haemodynamic task response, accounting for delay rather than assuming instant neural timing. BOLD is an indirect vascular proxy and is affected by motion and physiology; correlation is not direct measurement of neural firing.

Proposed allocation: Activation and oxygenation/deoxyhaemoglobin (1); Susceptibility and MR signal mechanism (1); Rapid EPI rationale (1); Repeated blocks / voxel association (1); Indirect/delayed or motion limitation (1). **Total 5.**

Source map: S4 pp.84-85,105-108; syllabus p.1. **Closest past question:** No direct historical fMRI question. **Status:** Original explicit-syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

3(f) [4 marks]

Question. Explain time-of-flight MR angiography in terms of inflowing blood and saturated stationary tissue. Why can a large tip angle relative to the stationary tissue Ernst angle improve vessel contrast, and what is one limitation?

Expected answer and working. Rapid repeated GE excitations partially saturate stationary spins. Fresh blood entering the slab has experienced fewer RF pulses and has more longitudinal magnetization, giving a stronger signal than stationary background. A relatively large angle further suppresses the stationary steady-state signal while fresh spins can still respond strongly. Slow flow, in-plane flow or repeated excitation through a thick slab can saturate blood too, reducing vessel brightness. The explanation concerns inflow history, not a universal intrinsic shortening of blood T1.

Proposed allocation: Stationary saturation (1); Fresh inflow signal (1); Relative tip-angle contrast reasoning (1); Slow/in-plane flow limitation (1). **Total 4.**

Source map: S4 pp.102-104. **Closest past question:** No direct historical MR-angiography question. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

Model 4 | Question 4: Ultrasound instrumentation, OCT and thermography

4(a) [4 marks]

Question. Explain piezoelectric transmission and reception in an ultrasound probe. Sketch a basic pulse-echo system signal path including a transmitter/receiver switch and display.

Expected answer and working. An applied alternating electric field causes the piezoelectric element to expand and contract, launching sound (inverse effect). Returning pressure deforms the element and produces an electrical response (direct effect). A transmit/receive switch protects and routes the receiver during high-voltage firing. Receive amplification, depth-dependent gain, envelope detection and scan conversion produce a brightness image. Coupling, timing and acoustic backing/focusing influence the response.



Receive path returns through T/R switch; schematic shows functional sequence

Proposed allocation: Inverse effect / transmission (1); Direct effect / reception (1); T/R switch and receive processing (1); Labeled signal path (1). **Total 4.**

Source map: S05 pp.17-21,40. **Closest past question:** 2024 Q4(a)iv is closest system context. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(b) [4 marks]

Question. Compare linear and phased arrays for beam formation. Explain how electronic focusing improves lateral resolution and give one cost of using several transmit focal zones.

Expected answer and working. A linear array typically uses successive subapertures to produce approximately parallel lines and a rectangular image. A phased array applies inter-element delays across an aperture to steer a sector beam. Curved delay profiles align wave arrival at a chosen focus, narrowing the beam and improving lateral resolution there; receive delays can vary with echo depth. Several transmit focal zones require additional firings and reduce frame rate at fixed depth and line count.

Proposed allocation: Linear subaperture scan (1); Phased steering delays (1); Focusing/narrower beam (1); Multiple-focus frame-rate cost (1). **Total 4.**

Source map: S05 pp.23-31. **Closest past question:** 2024 Q4(a)iv; 2025 Q4(a)iii. **Status:** Original syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(c) [3 marks]

Question. Explain what Doppler ultrasound adds to B-mode imaging. State how motion towards versus away from the probe affects the detected frequency, and why perpendicular flow is difficult to measure with this principle.

Expected answer and working. B-mode displays structure from echo strength; Doppler measures frequency shifts associated with moving scatterers, commonly blood, and provides flow-related functional information. Motion towards the probe raises the received frequency and motion away lowers it under the usual backscatter convention. The shift depends on the velocity component along the beam; perpendicular motion has approximately zero component and little Doppler shift.

Proposed allocation: Flow information beyond structure (1); Towards/away shift directions (1); Beam-axis component / perpendicular limitation (1). **Total 3.**

Source map: S05 pp.3,39; syllabus p.1. **Closest past question:** No direct historical Doppler question. **Status:** Original explicit-syllabus coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(d) [2 marks]

Question. State two physical reasons why ultrasound may fail to provide a useful image of a target behind an air-filled cavity or thick bone.

Expected answer and working. The large acoustic-impedance mismatch causes strong reflection and poor transmission through the interface. Bone and air also strongly attenuate the propagating beam, so target echoes can fall below the detector range and posterior acoustic shadowing results. Changing gain cannot reconstruct echoes that never reach the receiver.

Proposed allocation: Impedance/reflection barrier (1); Attenuation/weak deep echoes (1). **Total 2.**

Source map: S05 pp.4,13-16,38. **Closest past question:** 2025 Q4(a)i. **Status:** Recurring family.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(e) [6 marks]

Question. Compare time-domain and Fourier-domain OCT. Explain A-scan-to-B-scan formation, give one ophthalmic application, and identify one acquisition or processing error that could corrupt a thickness measurement.

Expected answer and working. Time-domain OCT scans reference delay mechanically to sample depth sequentially. Fourier-domain OCT obtains depth information from an interference spectrum: a spectrometer is used in spectral-domain systems, or a swept source plus detector in swept-source systems; Fourier processing recovers the depth profile. Lateral juxtaposition of A-scans forms a B-scan. Retinal-layer/nerve-fibre-layer monitoring is an appropriate application. Eye motion, poor focus, blink dropout or incorrect automated layer segmentation can corrupt measured thickness, so image and segmentation boundaries require review.

Proposed allocation: Time-domain reference-delay scan (1); Fourier-domain spectral acquisition (1); Fourier recovery of depth (1); A/B scan formation (1); Ophthalmic application (1); Error + effect on measurement (1). **Total 6.**

Source map: A2 PDF pp.3-8; syllabus p.1. **Closest past question:** 2024 Q4(b)ii; 2025 Q4(b)iii. **Status:** Historical basics plus assignment-supported coverage.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.

4(f) [6 marks]

Question. A thermal camera compares a body surface at 307 K with a reference surface at 305 K. Assuming equal emissivity and the supplied total-emission law $M=\varepsilon\sigma T^4$, calculate the percentage increase of emitted power. Explain why the change is not by itself a diagnosis, and give three controls for a reliable comparison.

Expected answer and working. Equal ε and σ cancel: $M_{307}/M_{305}=(307/305)^4=1.02648863$, so emitted power rises by about 2.64886%. This calculation describes total emitted radiant power, not a calibrated band-limited camera voltage directly. Temperature patterns are nonspecific and affected by environment and perfusion; a change alone cannot establish a particular disease. Controls include a calibrated reference/check of detector uniformity, stable room conditions with patient acclimatization, and consistent emissivity/view angle/distance/region positioning. Use thermal findings with clinical context.

Proposed allocation: T^4 ratio setup (1); 2.64886% result (1); Nonspecific / band-limited interpretation (1); Calibration control (1); Environmental/patient control (1); Geometry/emissivity control (1). **Total 6.**

Source map: A1 PDF pp.4-5,8-9; syllabus p.1. **Closest past question:** 2023 Q4(b)i is closest principle precedent. **Status:** Original assignment-supported numerical.

Mark-losing omissions: unsupported final numbers, missing units or assumptions in calculations; unlabelled sketches or lists without the requested physical explanation in theory parts.